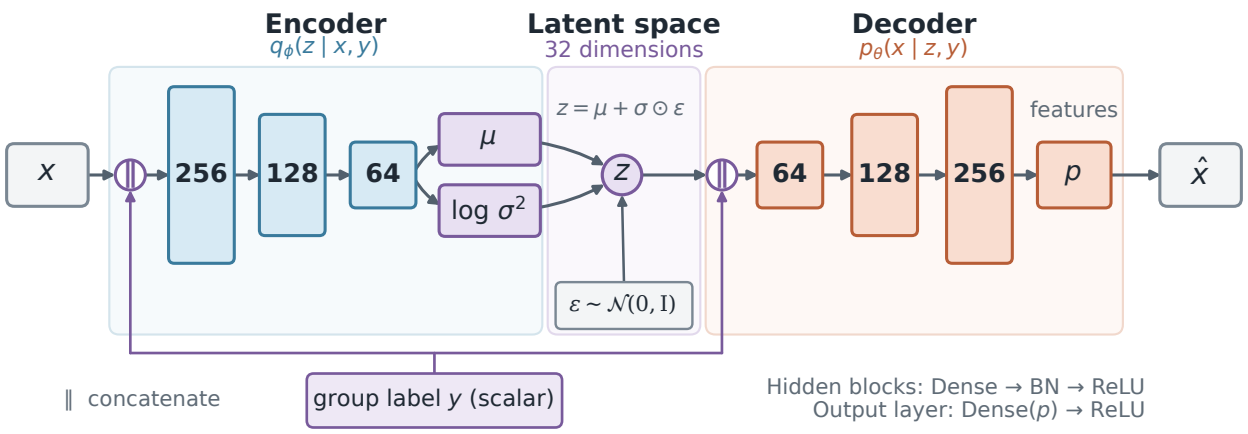
Encoder
$q_\phi(z \mid x, y)$
Latent space
32 dimensions
Decoder
$p_\theta(x \mid z, y)$
x
256
128
64
μ
$\log \sigma^2$
$z = \mu + \sigma \odot \varepsilon$
z
$\varepsilon \sim \mathcal{N}(0, \mathrm{I})$
64
128
256
p
features
$\hat{x}$
‖ concatenate
group label y (scalar)
Hidden blocks: Dense → BN → ReLU
Output layer: Dense(p) → ReLU